

Ordinary Differential Equations

ASSIGNMENT 4 – MA 417-2025

TOPIC : BOUNDARY VALUE PROBLEMS.

02.10.2025

1.1 Find the eigenvalues and eigenfunctions of the following boundary value problems:

(a) $y''(t) + \lambda y(t) = 0$, on $(0, 1)$, with boundary conditions $y(0) = 0$ and $y(1) + y'(1) = 0$.

(b) $y''(t) + \lambda y(t) = 0$, on $(0, 2\pi)$, with boundary conditions $y(0) = y(2\pi)$ and $y'(0) = y'(2\pi)$.

1.2 Show that the boundary value problem $y''(t) + \lambda y(t) = 0$ with $y(0) - y(\pi) = 0$ and $y'(0) + y'(\pi) = 0$ is not self-adjoint and every complex number is eigenvalue of this problem.

1.3 Consider the Sturm-Liouville differential equation

$$[p(x)y']' - q(x)y + \lambda r(x)y = 0, \quad 0 < x < 1,$$

where the functions $p(x)$, $p'(x)$, $q(x)$, and $r(x)$ are real-valued and continuous on $[0, 1]$. Assume that

$$p(x) > 0, \quad r(x) > 0 \quad \text{for all } x \in [0, 1].$$

The boundary conditions are

$$\alpha_1 y(0) + \alpha_2 y'(0) = 0, \quad \beta_1 y(1) + \beta_2 y'(1) = 0,$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2$ are real constants such that

$$\alpha_1^2 + \alpha_2^2 \neq 0, \quad \beta_1^2 + \beta_2^2 \neq 0.$$

(a) Check whether the boundary value problem is self-adjoint.

(b) Show that the eigenvalues λ of the boundary value problem are all real and distinct.

1.4 Reduce the BVP in self-adjoint BVP form:

(a) Hermite equation: $y''(t) - 2ty'(t) + \lambda y(t) = 0$ with $y(0) = 0$ and $y(1) = 0$.

(b) Euler equation: $t^2 y''(t) + ty'(t) + \lambda y(t) = 0$ with $y(1) = 0$ and $y(e) = 0$.

1.5 Consider the boundary value problem $y'' + \lambda y = 0$ in $[0, \pi]$ with the boundary condition

$$m_{11}y(0) + m_{12}y'(0) + n_{11}y(\pi) + n_{12}y'(\pi) = 0$$

and

$$m_{21}y(0) + m_{22}y'(0) + n_{21}y(\pi) + n_{22}y'(\pi) = 0,$$

where m_{ij} and n_{ij} are real constants such that $m_{11}m_{22} - m_{12}m_{21} \neq 0$ or $n_{11}n_{22} - n_{12}n_{21} \neq 0$.

(a) Show that this boundary value problem is self-adjoint if and only if the following condition is satisfied

$$m_{11}m_{22} - m_{12}m_{21} = n_{11}n_{22} - n_{12}n_{21}.$$

(b) Under the same condition, show that the problem has no complex eigenvalues.

1.6 Find the appropriate Green's function and express the solution of the following boundary value problem as a definite integral:

$$-(y'' + y) = 1; \quad y(0) = 0 \text{ and } y(1) = 1.$$

1.7 Consider the regular Sturm–Liouville boundary value problem:

$$L[y](t) = -\frac{d}{dt} \left(p(t) \frac{dy}{dt} \right) + q(t)y = f(t), \quad t \in [0, 1],$$

where

- $p \in C^1([0, 1])$ with $p(t) > 0$ for all $t \in [0, 1]$,
- $q, f \in C([0, 1])$.

The problem is subject to the boundary conditions:

$$U_1(y) = \alpha_1 y(0) + \alpha_2 y'(0) = 0, \quad U_2(y) = \beta_1 y(1) + \beta_2 y'(1) = 0,$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{R}$ satisfy

$$\alpha_1^2 + \alpha_2^2 \neq 0, \quad \beta_1^2 + \beta_2^2 \neq 0.$$

Assume that $\lambda = 0$ is **not** an eigenvalue of the homogeneous problem

$$L[y] = \lambda y$$

with the same boundary conditions $U_1(y) = 0$ and $U_2(y) = 0$.

(a) Prove that there exists a Green's function $G(t, s)$ such that the solution φ to the inhomogeneous problem can be expressed as

$$\varphi(t) = \int_0^1 G(t, s) f(s) ds.$$

Moreover, show that $G(t, s)$ is given by

$$G(t, s) = \begin{cases} -\frac{y_1(s)y_2(t)}{p(t)W(y_1, y_2)(t)}, & 0 \leq s \leq t \leq 1, \\ -\frac{y_1(t)y_2(s)}{p(s)W(y_1, y_2)(s)}, & 0 \leq t \leq s \leq 1, \end{cases}$$

where

- y_1, y_2 are non-trivial solutions of $L[y] = 0$ satisfying the boundary conditions at $t = 0$ and $t = 1$ respectively,
- $W(y_1, y_2)(t)$ is the Wronskian of y_1 and y_2 .

(b) Show that the Green's function satisfies the jump condition in its partial derivative at $t = s$:

$$\lim_{t \rightarrow s^+} \frac{\partial}{\partial t} G(t, s) - \lim_{t \rightarrow s^-} \frac{\partial}{\partial t} G(t, s) = -\frac{1}{p(s)}.$$

(c) Using the fact that $p(t)W(y_1, y_2)(t)$ is constant on $[0, 1]$, prove that the Green's function G is symmetric; that is,

$$G(t, s) = G(s, t).$$